

Практическая работа №1

В процессе выполнения практической работы были использованы ОС Fedora Linux, терминалы GNOME Ptyxis и VS Code Terminal.

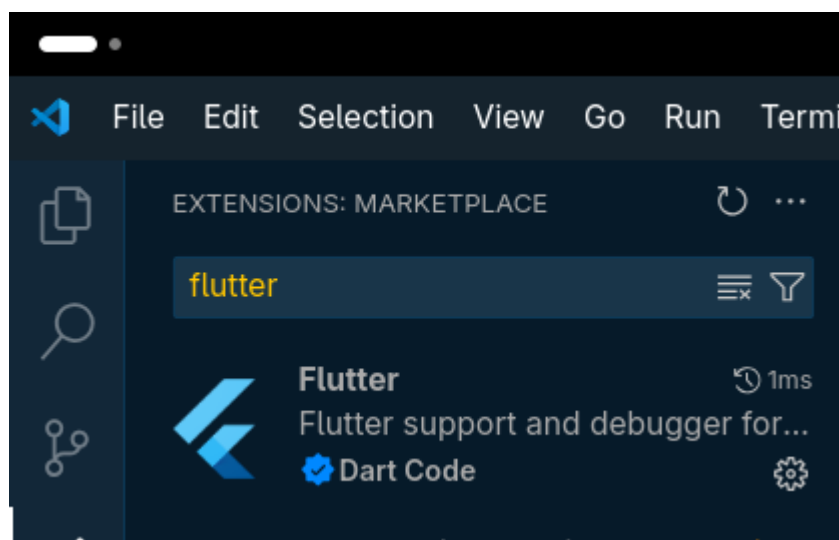
Основные этапы выполнения работы:

```
drozdov@fedora:~$ sudo dnf install -y curl git unzip xz zip mesa-libGLU
Обновление и загрузка репозитория:
Репозитории загружены.
Пакет "curl-8.11.1-5.fc42.x86_64" уже установлен.
Пакет "git-2.51.0-2.fc42.x86_64" уже установлен.
Пакет "unzip-6.0-66.fc42.x86_64" уже установлен.
Пакет "xz-1:5.8.1-2.fc42.x86_64" уже установлен.
Пакет "zip-3.0-43.fc42.x86_64" уже установлен.
Пакет "mesa-libGLU-9.0.3-6.fc42.x86_64" уже установлен.
```

Нечего делать.

```
drozdov@fedora:~$ sudo dnf install glibc libstdc++ bzip2-libs zlib
Обновление и загрузка репозитория:
Репозитории загружены.
Пакет "glibc-2.41-11.fc42.x86_64" уже установлен.
Пакет "libstdc++-15.2.1-1.fc42.x86_64" уже установлен.
Пакет "bzip2-libs-1.0.8-20.fc42.x86_64" уже установлен.
Пакет "zlib-ng-compat-2.2.5-2.fc42.x86_64" уже установлен.
```

```
drozdov@fedora:~$ groups
drozdov wheel input libvirt
drozdov@fedora:~$
```



Which Flutter template?

Application

Applications

A Flutter application with descriptive comments and tests.

Empty Application

A Flutter application without descriptive comments or tests.

Module

Other Project Types

A project to add a Flutter module to an existing Android or iOS applica...

Package

A shareable Flutter project containing modular Dart code.

Plugin

A shareable Flutter project containing an API in Dart code with a platform-specific implem...

```
[x] Android toolchain - develop for Android devices [13ms]
  x Unable to locate Android SDK.
    On this platform, it will assist you in installing the Android SDK components.
    (or visit https://flutter.dev/to/linux-android-setup for detailed instructions).
    If the Android SDK has been installed to a custom location, please use
    `flutter config --android-sdk` to update to that location.

[x] Chrome - develop for the web (Cannot find Chrome executable at google-chrome) [6ms]
  ! Cannot find Chrome. Try setting CHROME_EXECUTABLE to a Chrome executable.

[x] Linux toolchain - develop for Linux desktop [87ms]
  x clang++ is required for Linux development.
    It is likely available from your distribution (e.g.: apt install clang), or can be downloaded from https://releases.llvm.org/
  x CMake is required for Linux development.
    It is likely available from your distribution (e.g.: apt install cmake), or can be downloaded from https://cmake.org/download/
  • ninja version 1.12.1
  • pkg-config version 2.3.0
  ! Unable to access driver information using 'eglinfo'.
  ! It is likely available from your distribution (e.g.: apt install mesa-utils)

[!] Android Studio (not installed) [5ms]
  • Android Studio not found; download from https://developer.android.com/studio/index.html
    (or visit https://flutter.dev/to/linux-android-setup for detailed instructions).

[✓] VS Code (version 1.103.2) [5ms]
  • VS Code at /usr/share/code
  • Flutter extension version 3.118.0

[✓] Connected device (1 available) [15ms]
  • Linux (desktop) • linux • linux-x64 • Fedora Linux 42 (Workstation Edition) 6.16.3-200.fc42.x86_64

[✓] Network resources [271ms]
  • All expected network resources are available.

! Doctor found issues in 5 categories.
exit code 0
```

```
drozdov@fedora:~$ echo 'export PATH="$PATH:/home/drozdov/flutter/bin"' >> ~/.bashrc
drozdov@fedora:~$ source ~/.bashrc
drozdov@fedora:~$ flutter -h
Manage your Flutter app development.
```

Common commands:

```
flutter create <output directory>
    Create a new Flutter project in the specified directory.

flutter run [options]
    Run your Flutter application on an attached device or in an emulator.
```

Usage: flutter <command> [arguments]

Global options:

```
-h, --help          Print this usage information.
-v, --verbose       Noisy logging, including all shell commands executed.
                    If used with --help, shows hidden options. If used with
```

```
drozdov@fedora:~$ flutter upgrade
Flutter is already up to date on channel stable
Flutter 3.35.3 • channel stable • https://github.com/flutter/flutter.git
Framework • revision a402d9a437 (22 часа назад) • 2025-09-03 14:54:31 -0700
Engine • hash 672c59cfa87c8070c20ba2cd1a6c2a1baf5cf08b (revision ddf47dd3ff) (23 hours ago) •
2025-09-03 20:02:13.000Z
Tools • Dart 3.9.2 • DevTools 2.48.0
drozdov@fedora:~$
```



Downloading Components

```
Installing Android SDK Platform 36 in /home/drozdov/Android/Sdk/platforms/android-36
"Install Android SDK Platform 36 (revision 2)" complete.
"Install Android SDK Platform 36 (revision 2)" finished.
Preparing "Install Android SDK Build-Tools 36 v.36.0.0".
Downloading https://dl.google.com/android/repository/build-tools_r36_linux.zip
"Install Android SDK Build-Tools 36 v.36.0.0" ready.
Installing Android SDK Build-Tools 36 in /home/drozdov/Android/Sdk/build-tools/36.0.0
"Install Android SDK Build-Tools 36 v.36.0.0" complete.
"Install Android SDK Build-Tools 36 v.36.0.0" finished.
Preparing "Install Android SDK Platform-Tools v.36.0.0".
Downloading https://dl.google.com/android/repository/platform-tools_r36.0.0-linux.zip
"Install Android SDK Platform-Tools v.36.0.0" ready.
Installing Android SDK Platform-Tools in /home/drozdov/Android/Sdk/platform-tools
"Install Android SDK Platform-Tools v.36.0.0" complete.
"Install Android SDK Platform-Tools v.36.0.0" finished.
SDK Manager found the following installed packages: build-tools;36.0.0 build-tools;36.1.0-rc1
emulator platform-tools platforms;android-36 sources;android-36 system-images;android-36;
google_apis_playstore;x86_64
Android SDK is up to date.
```

Previous

Next

Cancel

Finish

Add Device

Configure virtual device

Device

Additional settings

Name

Pixel 9

Select system image

Use the filters to help find the system image that you prefer. The combination of device profile and system image is only an approximation of the equivalent physical hardware.

API

Services

API 36.0 "Baklava"; Android 16.0

Google Play Store

System Image	API
Google Play Intel x86_64 Atom System Image	36.0
Pre-Release 16 KB Page Size Google Play Intel x86_64 Atom System Ima...	36.0

☐ Show system images with SDK extensions
 ☐ Show unsupported system images

Pixel 9

Device

OEM

Supported API Levels

Device

System Image

Screen

Google

35+

36.0

Google Play

x86_64

arm64-v8a

1080 × 2424

420 dpi

```

• drozdov@fedora:~/mirea/pks/hello_app$ flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.35.3, on Fedora Linux 42 (Workstation Edition) 6.16.3-200.fc42.x86_64, locale ru_RU.UTF-8)
[✓] Android toolchain - develop for Android devices (Android SDK version 36.1.0-rc1)
[✓] Chrome - develop for the web
[✓] Linux toolchain - develop for Linux desktop
    ! Unable to access driver information using 'eglinfo'.
      It is likely available from your distribution (e.g.: apt install mesa-utils)
[✓] Android Studio (version 2025.1.3)
[✓] VS Code (version 1.103.2)
[✓] Connected device (2 available)
[✓] Network resources

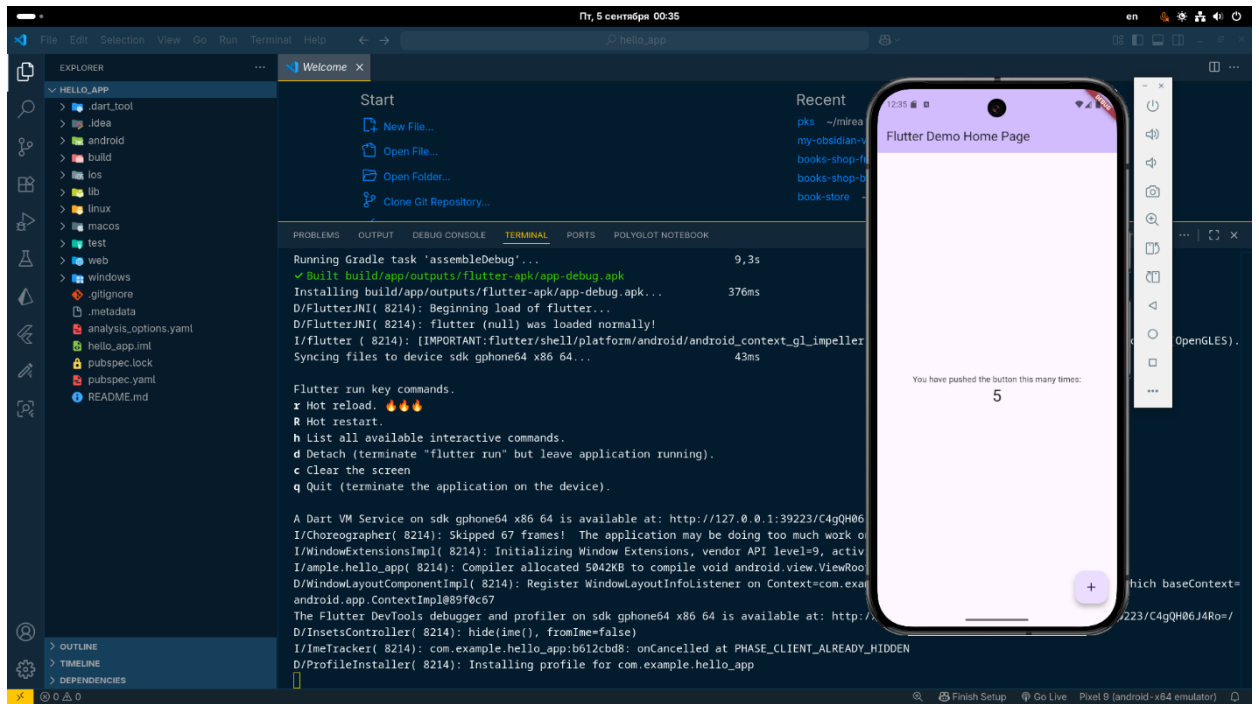
• No issues found!
• drozdov@fedora:~/mirea/pks/hello_app$
  
```

```

• drozdov@fedora:~/mirea/pks/hello_app$ flutter devices
Found 3 connected devices:
sdk gphone64 x86 64 (mobile) • emulator-5554 • android-x64 • Android 16 (API 36) (emulator)
Linux (desktop)           • linux        • linux-x64   • Fedora Linux 42 (Workstation Edition) 6.16.3-200.fc42.x86_64
Chrome (web)              • chrome       • web-javascript • Google Chrome 139.0.7258.154

Run "flutter emulators" to list and start any available device emulators.

If you expected another device to be detected, please run "flutter doctor" to diagnose potential issues. You may also try increasing the time to wait for connected devices with the "--device-timeout" flag. Visit https://flutter.dev/setup/ for troubleshooting tips.
  
```



Проблемы и способы их решения:

Не были установлены некоторые элементы из списка doctor summary. Для решения проблемы были загружены браузер Chrome, а также некоторые C++ пакеты по типу clang++, cmake и т.д.